logistics

main textbooks (1)

Peterson and Davie, Computer Networks: A Systems Approach

- available online https://book.asystemsapproach.org/, also paper version
- focused on higher-level goals, not details of specific implementations

Fall and Stevens, TCP/IP Illustrated: Volume 1, Second Edition: The Protocols

- available via O'Reilly through library
  - link on schedule page, or search library catalog
  - make account with your UVa email
- focused on details of the Internet and related systems

textbooks (2)

Dordal, An Introdcution to Computer Networks

- secondary textbook
- https://intronetworks.cs.luc.edu/
- more coverage of low-level wireless, ethernet

coursework

weekly quizzes

- approx 4 questions/week
- mostly autograded, will manually adjust based on work/comments
- 20% of overall course grade, lowest dropped automatically

homework assignments

- mostly programming-based, mostly Python
- approx. weekly deadlines
- 60% of overall course grade

final exam

- 16 December
- 20% of overall course grade

homeworks preview (1)

framing — transmit messages using bits in Python

reliable1/2 — stream of messages with automatic resending

P4* — forwarding messages locally

AIMD* — simulating variations on TCP

packet scheduling — prioritizing network traffic

DV routing — forwarding messages far

DNS — mapping names to addresses

webserver

wireless*

assignment infrastructure

most assignments in Python

P4 — requires some VM setup

- network/switch emulation software that really needs root on Linux

AIMD — using C++ simulator

- goal: tweak and measure working TCP implementation
- mostly small modifications, shouldn't need much C++ knowledge
- (the other obvious simulator is mainly in Tcl, which seems worse idea)

notable changes

AIMD assignment is new

- previous assignment was implementing simple congestion control
- (congestion control = avoiding overloading network)
- too many corner cases
- did not complement lecture material on modern congestion control

framing assignment has new test program

maybe other changes, time-depending

- would like to rework wireless assignment

due times/late policy

assignments usually due Friday @ 11:59pm

- 90% credit for submissions up to 72 hours late, no questions asked

quizzes usually due Tuesday @ 12:15pm

- done online, untimed, open notes/book
- late submissions not normally accepted (may discuss during lecture)

unusual circumstances — contact me

- can often grant exceptions/alternate arrangements
- more flexibility if contacted earlier

collaboration

quizzes, homeworks are individual

yes:

- review examples in textbook/lecture/etc.
- discuss high-level concepts (without detailed psuedocode for assignment, etc.)
- lookup up syntax/standard library documentation

no:

- sharing code/detailed psuedocode/answers
- getting code/etc. from web forums, LLMs, etc.

gen AI

rule of thumb: can ask things you could ask fellow students .5cm

no: having LLM write your code, your writeup, …

aside: selected standard python libraries

bisect — search/insert within sorted list

collections's deque — ordered list with fast insertion/deletion at both ends

heapq — priority queue (using a heap)

struct — convert between integers and bytes and vice-versa

array — convert between integers and bytes in less flexible way

not intending for you to reimplement common, non-networking tasks

may grant exceptions to use other libraries/etc. for generic tasks

lectures

will attempt to record (but sometimes technical stuff happens)

- will link recordings from schedule page

will not take attendance

homework/topic sync

rough plan on schedule page

imagine things might get out of sync